

Math 465: Homework 5

Due: Thursday, October 8, by 11:59PM

Instructions. Write full and complete solutions to the following problems and submit them on Gradescope by the deadline. For each problem, explain the reasoning behind your answer; a formula or numerical answer without justification is not a complete solution. **If your final answer is numerical, please carry out the arithmetic** and state the resulting number (so, for example, if your answer is $6! - 5!$, I want you to also tell me the answer is 600.)

You may discuss the problems with other students, but you must write up your solutions independently and in your own words. You are not allowed to use generative AI tools, Internet searches, or any other external means to directly obtain a solution. Generative AI tools may also not be used to proofread your writing or L^AT_EX. If you are unsure about what is or is not permitted, please ask.

At the beginning of your submission, acknowledge any permitted books, notes, or other sources you consulted, and list the names of any students with whom you discussed the homework.

1. A polynomial $p(x)$ of degree at most 4 satisfies

$$p(0) = 1, \quad p(2) = 4, \quad p(4) = 13, \quad p(6) = 40, \quad p(8) = 109.$$

- (a) Use finite differences to find $p(10)$ and $p(-2)$.
- (b) Find a formula for $p(x)$ and explain why it is uniquely determined. You may leave your answer as a linear combination of binomial coefficients $\binom{x/2}{k}$.

2. Let $d \geq 0$ be an integer, and let $p(x)$ be a real polynomial of degree at most d .

- (a) Prove that if $p(0), p(1), \dots, p(d)$ are integers, then $p(m)$ is an integer for every integer m , including negative integers.
- (b) Does this imply, in general, that all coefficients of $p(x)$ in the ordinary power basis $1, x, x^2, \dots$ are integers? Prove your answer or give a counterexample.

3. You may use the identity $x^m = \sum_{k=0}^m S(m, k)(x)_k$, where $S(m, k)$ is a Stirling number of the second kind and $(x)_k = x(x-1) \cdots (x-k+1)$, with $(x)_0 = 1$.

- (a) For integers $m \geq 1$ and $N \geq 0$, prove that

$$\sum_{j=0}^{N-1} j^m = \sum_{k=0}^m k! S(m, k) \binom{N}{k+1}.$$

An empty sum is understood to be 0.

- (b) Use this identity to express $\sum_{j=0}^{N-1} j^4$ as a linear combination of binomial coefficients. Evaluate all Stirling numbers and factorials appearing in the coefficients.

4. Consider the recurrence

$$h_n - 3h_{n-1} + 2h_{n-2} = 2^n \quad (n \geq 2).$$

- (a) Show that no sequence of the form $h_n = A2^n$, with A constant, satisfies this recurrence.
- (b) Find a closed formula for the solution satisfying $h_0 = 0$ and $h_1 = 1$.

5. Let $n \geq 1$ and $0 \leq r \leq n$ be integers. Consider all paths from $(0, 0)$ to $(2n, 0)$ consisting of steps $(1, 1)$ and $(1, -1)$.

- (a) How many of these paths stay entirely in the region $y \geq -r$?
- (b) How many have minimum height exactly $-r$?

Give formulas in terms of binomial coefficients and justify them using the reflection principle. Use the convention that $\binom{a}{b} = 0$ for nonnegative integers a when $b < 0$ or $b > a$.

6. For $n \geq 1$, let t_n be the number of triangulations of a fixed convex polygon with vertices $v_0, v_1, \dots, v_{n+1}$ in cyclic order. A triangulation is a subdivision of the polygon into triangles by noncrossing diagonals. Triangulations that differ by rotation are considered distinct. Set $t_0 = 1$.

- (a) By considering the triangle incident to the edge v_0v_{n+1} , derive a recurrence for t_n . Use it to prove that $t_n = C_n$, where C_n is the n th Catalan number.
- (b) For $n \geq 2$ and $1 \leq r \leq n - 1$, how many triangulations contain the diagonal v_0v_{r+1} ? Give your answer in terms of Catalan numbers.

7. For $n \geq 1$, a *fully bracketed expression* (or *bracketing*) on $a_1, \dots, a_{n+1}$ specifies how to combine these letters using a binary operation, which takes exactly two inputs at a time, while keeping the letters in their original order. We write the operation by juxtaposition: combining two expressions U and V produces (UV) , where each of U and V is either a single letter or an already fully bracketed expression. Include one pair of parentheses for every application of the operation, including the outermost pair.

For example,

$$((a_1a_2)(a_3a_4))$$

is a bracketing: combine a_1 with a_2 , a_3 with a_4 , and then combine the two results. In contrast,

$$((a_1a_2)a_3a_4)$$

is not: the outermost parentheses contain three inputs. Also, $(a_1a_2)(a_3a_4)$ is not fully bracketed under our convention because the outermost pair of parentheses is missing.

- (a) Consider the construction that deletes all letters and replaces opening parentheses by $+$ and closing parentheses by $-$. Show that this construction is not injective in general by giving two distinct bracketings of the same sequence of letters with the same image.
- (b) Consider instead the following construction: delete all closing parentheses and the final letter a_{n+1} ; replace each opening parenthesis by $+$ and each remaining letter by $-$. Prove that the resulting word is a ballot sequence of length $2n$.
- (c) Describe an inverse to the construction in part (b), prove that it is an inverse, and deduce the number of fully bracketed expressions on $n + 1$ letters.

8. A *peak* of a Dyck path is an up-step immediately followed by a down-step. Let $\mathcal{D}_n$ be the set of Dyck paths of length $2n$, using steps $(1, 1)$ and $(1, -1)$ and staying in $y \geq 0$. Define

$$P_n(q) = \sum_{D \in \mathcal{D}_n} q^{\text{pk}(D)}, \quad P_0(q) = 1,$$

where $\text{pk}(D)$ is the number of peaks of D .

- (a) Compute $P_1(q)$, $P_2(q)$, and $P_3(q)$.
(b) Prove that, for $n \geq 0$,

$$P_{n+1}(q) = qP_n(q) + \sum_{k=1}^n P_k(q)P_{n-k}(q).$$

- (c) For $n \geq 1$, associate a Dyck path to an ordered rooted tree on $n + 1$ unlabeled vertices as follows: first use the bijection from class to obtain a ballot sequence, then interpret each $+$ as an up-step and each $-$ as a down-step. Under this correspondence, what does the number of peaks of the Dyck path count in the tree? Explain your answer.

9. A permutation $\sigma \in S_n$ is *132-avoiding* if there are no indices $i < j < k$ such that

$$\sigma(i) < \sigma(k) < \sigma(j).$$

Recall that its Rothe diagram is

$$D(\sigma) = \{(i, j) \in \{1, \dots, n\}^2 : j < \sigma(i) \text{ and } i < \sigma^{-1}(j)\}.$$

Rows are numbered from top to bottom, and columns from left to right. Let d_i be the number of cells in row i of $D(\sigma)$.

- (a) Prove that σ is 132-avoiding if and only if each row of $D(\sigma)$ consists of its leftmost d_i cells and

$$d_1 \geq d_2 \geq \dots \geq d_n \geq 0.$$

Empty rows are allowed. Also show that $d_i \leq n - i$ for every i .

- (b) Use Rothe diagrams to give an explicit bijection between 132-avoiding permutations in S_n and Dyck paths of length $2n$. Explain how to recover the permutation from the path, and deduce the number of 132-avoiding permutations in S_n .

10. For $n \geq 1$, let

$$Q_n(x_1, \dots, x_n) = \prod_{i=1}^n (x_1 + \dots + x_i).$$

We count distinct monomials after collecting like terms, without regard to their coefficients.

- (a) Let $b_1, \dots, b_n$ be nonnegative integers with sum n . Prove that the monomial $x_1^{b_1} \dots x_n^{b_n}$ occurs in Q_n if and only if

$$b_1 + \dots + b_j \geq j \quad \text{for every } 1 \leq j \leq n.$$

- (b) For $1 \leq r \leq n$, count the distinct monomials in Q_n in which the exponent of x_1 is exactly r . Give an answer in terms of binomial coefficients.

Hint: Encode the exponent vector as a ballot sequence, and consider its initial run of plus signs.